7.3.7 Language analyzers

Elasticsearch provides language analyzers for working with the following languages: Arabic, Armenian, Basque, Bengali, Bulgarian, Catalan, Czech, Dutch, English, Finnish, French, Galician, German, Hindi, Hungarian, Indonesian, Irish, Italian, Latvian, Lithuanian, Norwegian, Portuguese, Romanian, Russian, Sorani, Spanish, Swedish, and Turkish. We can configure these out-of-the-box language analyzers to add a stopwords filter so we don't index a language's unnecessary (or common) words. The following listing demonstrates three language analyzers (English, German, and Hindi) in action.

Listing 7.18 English, German, and Hindi language analyzers

```
POST _analyze
{
  "text": "She sells Sea Shells on the Sea Shore",
  "analyzer": "english"
}
```

Uses the english language analyzer. The output is ["she", "sell", "sea", "shell", "sea", "shore"].

```
POST _analyze
{
  "text": "Guten Morgen",
  "analyzer": "german"
}
```

Uses the german language analyzer. The output is ["gut","morg"].

```
POST _analyze
{
  "text": "नमस्ते कैसी हो तुम",
  "analyzer": "hindi"
}
```

Uses the hindi language analyzer. The output is ["नमस्ते","कैसी","तुम"].

We can configure a language analyzer with a few additional parameters to provide our own list of stop words or ask the analyzer to exclude the stemming operation. For example, the `stop` token filter used by the `english` analyzer categorizes a handful of words as stop words. We can override this list based on our requirements. For example, if we only want to override *a*, *an*, *is*, *and*, and *for*, we can configure our stop words as the following listing shows.

Listing 7.19 Index with custom stop words on the `english` analyzer

```
PUT index_with_custom_english_analyzer
{
  "settings": {
    "analysis": {
      "analyzer": {
        "index_with_custom_english_analyzer":{
          "type":"english",
          "stopwords":["a","an","is","and","for"]
        }
```

Creates an index with analyzer settings

Provides a custom name

The type of analyzer here is english.

Provides our set of stop words